

OMNIQ • Presentation Content Plan

16-Slide Competition Pitch — Qilu 2026

SLIDE 01 • COVER

OMNIQ *The AI Operations Layer for the Factory Floor*

See Everything • Know Everything • Miss Nothing

Team: AXIOM — Tongmyong University BIC

SLIDE 02 • THE SCALE OF THE PROBLEM

Every year, the global manufacturing industry loses **\$50 billion** to unplanned downtime. The average factory experiences **800 hours** of equipment downtime per year — more than 15 hours every single week. Industrial accidents cost billions more. And **97% of manufacturers** are actively worried about losing critical knowledge when their expert workers retire.

This is not a technology problem. The technology to fix all of this already exists. So why is it still happening?

SLIDE 03 • THREE DAILY FAILURES EVERY FACTORY FACES

Failure 1: The Unseen Accident A worker walks into a restricted zone without a hard hat. No supervisor saw it. The injury report is filed after the fact — not before. Manual safety monitoring is reactive, inconsistent, and human. Humans get tired. AI does not.

Failure 2: The Retiring Expert Your best engineer retires on Friday. He spent 22 years learning the quirks of every machine on the floor — the workarounds, the failure patterns, the shortcuts nobody wrote down. When he leaves, that knowledge leaves with him. A 500-page manual doesn't cover any of it. The junior worker who replaces him has no one to call at 2am when something breaks.

Failure 3: The Missed Defect A human inspector catches roughly 80% of defects. The other 20% ships to customers — returns, complaints, lost contracts. The solution exists: AI vision inspection systems that catch 99.9% of defects. But they cost \$50,000 or more. 90% of small and medium factories cannot afford them. So they keep using human eyes and accepting 20% misses.

SLIDE 04 • THE REAL PROBLEM IS ACCESS, NOT TECHNOLOGY

The large enterprise factories — the ones that supply the big brands — have already solved these problems. They have AI safety monitoring, automated quality inspection, digital documentation systems. They spent millions to get there.

The 50,000 small and medium factories in Shandong Province alone don't have any of it. Not because the technology doesn't exist. Because no one built it for them at a price they can afford.

A Siemens industrial AI system: tens of thousands of dollars per year, requires Siemens hardware, requires Siemens engineers.

What an SME factory in Yantai can afford: around **\$200 per month**.

That gap — between what exists and what's accessible — is the entire business.

SLIDE 05 • INTRODUCING OMNIQ

OMNIQ is an AI operations platform built specifically for small and medium factories in Asia. It gives any factory — regardless of size, brand, or budget — three things they've never had before:

- **AI eyes** that watch the factory floor 24/7
- **AI intelligence** that catches defects before they ship
- **AI knowledge** that knows every machine inside out

No expensive hardware. No specialist engineers. No vendor lock-in. Plug in any camera. Upload any manual. Start on day one.

One platform. Any factory. \$199 per month.

SLIDE 06 • THREE AGENTS, ONE PLATFORM

OMNIQ is built around three AI agents that work independently or together:

Agent 1 — Safety-Eye Real-time visual monitoring of the factory floor for safety compliance and zone violations.

Agent 2 — Visual-Check Automated quality inspection that any worker can set up in 60 seconds using only a phone camera.

Agent 3 — Expert-Flow An AI that reads every machine manual, SOP, and maintenance log — and answers any worker's question instantly, in their language, with citations.

Together, they cover the three most expensive daily failure points in any factory.

SLIDE 07 · SAFETY-EYE — DEEP DIVE

The problem it solves: Industrial accidents kill over 2.3 million workers globally every year. The vast majority are preventable. Most factories rely on periodic manual supervisor walkthroughs to check compliance — a method that is slow, inconsistent, and impossible to scale across multiple zones simultaneously.

What Safety-Eye does: It turns any existing IP camera or webcam into an AI safety monitor. The system watches every person on the floor in real time. It detects whether each person is wearing the required PPE — hard hat, safety vest, gloves. It detects when anyone enters a restricted or hazardous zone without authorization. The moment a violation is detected, a real-time alert fires to the supervisor's phone with a timestamped photo of the incident.

Why it matters: Every alert that prevents an accident also prevents insurance claims, production stoppages, regulatory penalties, and human suffering. Factories that deploy automated safety monitoring report significant reductions in incident rates within months. And the entire system runs on cameras the factory already has.

Key image context for AI generation: A factory floor camera feed with detection bounding boxes around workers — one green (compliant, hard hat visible), one red (violation, no hard hat detected). Alert notification visible. Industrial setting, realistic.

SLIDE 08 · VISUAL-CHECK — DEEP DIVE

The problem it solves: Quality defects that escape the production line cost manufacturers billions in returns, recalls, and lost customer relationships. Human visual inspection — the standard in most SME factories — catches only about 80% of defects. The other 20% ships. Enterprise AI inspection systems solve this problem with 99.9% accuracy, but they require custom lighting rigs, dedicated hardware, and upfront investments of \$50,000 or more. For a small factory making \$2 million in annual revenue, that's not a purchase — that's a risk.

What Visual-Check does: A worker takes 5 to 10 photographs of a "good" product — one with no defects, representing the correct standard. The AI learns what normal looks like from those images alone. From that point forward, any new product photographed by the same worker is automatically classified as Pass or Fail, with a visual heatmap showing exactly where the anomaly or defect is located. No labels needed. No data scientists. No expensive hardware. Just a phone camera and 60 seconds of setup.

Why it matters: This is the democratization of quality control. The same inspection capability that costs \$50,000 at the enterprise level is now available to any factory at a fraction of the cost. A small factory in Shandong making metal parts, a food packaging line in Korea, a textile factory in

Vietnam — all of them can now achieve the same defect detection accuracy as their largest competitors.

Key image context for AI generation: A factory worker holding a phone above a metal component. Phone screen shows an AI detection interface with a red heatmap highlight showing a scratch or defect. Realistic, hands-on, no expensive equipment visible.

SLIDE 09 • EXPERT-FLOW — DEEP DIVE

The problem it solves: The manufacturing industry is facing a knowledge crisis. An entire generation of expert engineers and technicians is retiring, taking decades of accumulated machine knowledge with them. Surveys show that only 30% of manufacturers consistently capture knowledge from departing employees. 41% rarely or never attempt it at all. What's left behind are machine manuals — often 300 to 500 pages long, written in Japanese, German, or English, dense with technical jargon — and junior workers who don't know where to start.

The result: machines sit idle while workers search for answers. Wrong repairs get made by technicians guessing from incomplete information. Training new workers takes months. And the expertise that made the factory work disappears, permanently.

What Expert-Flow does: Any document — machine manuals, standard operating procedures, maintenance logs, safety guidelines — gets uploaded to OMNIQ once. From that point, any worker on any shift can ask questions in plain language, in Korean, Chinese, or English, and receive an immediate, accurate answer with the exact source page cited. The AI does not guess. It retrieves. It explains. It teaches.

Ask: *"How do I clear an AL.E6 encoder error on Machine 3?"* Get back: a step-by-step resolution with the manual page number, the safety precautions to take, and a follow-up offer to explain further.

Why it matters: The average downtime event in manufacturing now lasts 81 minutes — up from 49 minutes five years ago — because the people who knew how to fix things fast are leaving. Expert-Flow gives every junior worker the knowledge of the most experienced engineer, available 24 hours a day, in their own language.

Key image context for AI generation: An older experienced factory engineer sitting across from a young worker, pointing at a machine while the young worker holds a tablet showing a chat interface with a technical answer. A mentorship moment made permanent by AI.

SLIDE 10 • HOW IT ALL WORKS TOGETHER

OMNIQ sits on top of what a factory already has. It does not require replacing any existing

equipment, hardware, or software. It connects to existing cameras. It reads existing documents. It works on any device with a browser.

The three agents share a common platform and a common dashboard. A plant manager can see, in one view: live safety compliance across all camera feeds, today's quality inspection pass/fail rates, and a log of every question workers have asked the AI that day. Every alert, every defect, every knowledge gap — visible, tracked, and actionable.

For factories that want to start small: begin with just Expert-Flow. Upload one machine manual. Let workers experience the difference. Then add Visual-Check for the highest-risk production line. Then add Safety-Eye cameras one at a time. The platform grows with the factory's confidence and budget.

Key image context for AI generation: A clean dashboard interface on a large monitor showing three panels: a live camera feed with AI overlays, a quality inspection results grid, and a chat interface. Dark, premium, unified. The impression of a control room for factory intelligence.

SLIDE 11 • WHY WE CAN BUILD THIS — THE DEMOCRATIZATION THESIS

The technology behind OMNIQ is not proprietary or experimental. It is built on open-source foundations that power enterprise AI systems costing millions of dollars:

- Safety-Eye uses **YOLOv8** — the same computer vision architecture used in autonomous vehicles and military surveillance systems. Pretrained datasets for PPE detection already exist and are publicly available.
- Visual-Check uses **Anomalib** with **DINOv2** — Intel's open-source industrial anomaly detection library, designed specifically for factory quality inspection. It requires only a handful of good-sample images to work.
- Expert-Flow uses **LlamaIndex** and **ChromaDB** — the same retrieval-augmented generation infrastructure used by enterprise knowledge management systems at Fortune 500 companies.

What took enterprise companies years and millions to build, we deploy in weeks using these foundations.

The democratization angle: Lovable made it possible for anyone to build a web app. GitHub Copilot made it possible for any developer to write code faster. OMNIQ is the same moment for industrial AI — taking powerful technology that existed only in enterprise environments and making it available to every factory that needs it.

We are not inventing new AI. We are removing the barriers that kept existing AI out of reach for 95% of the factories that need it most.

SLIDE 12 • THE MARKET OPPORTUNITY

The numbers:

- 50,000+ SME manufacturers in Shandong Province alone
- China has approximately 4 million manufacturing SMEs nationally
- The global industrial AI market is projected to reach \$47 billion by 2028
- Currently, over 95% of Asian SME factories have zero AI operational tools

The specific opportunity: Shandong's government has committed to building 30,000 smart factories under the national AI+ Manufacturing plan. The policy exists. The intent exists. The funding exists. What doesn't exist yet is an affordable, accessible platform that actually serves these factories at their scale.

OMNIQ is not trying to compete in the enterprise market. The enterprise market is owned by Siemens, SAP, and Rockwell — companies with hundreds of billions in combined market cap. OMNIQ's market is the enormous, underserved layer beneath them: the SME factories that make up the backbone of manufacturing in Korea, China, and across Asia.

At 1,000 factories paying \$199/month, OMNIQ generates \$2.4 million in annual recurring revenue. At 10,000 factories — still a fraction of the addressable market — that's \$24 million ARR with gross margins exceeding 90%.

SLIDE 13 • BUSINESS MODEL

How OMNIQ makes money: Subscription SaaS. Monthly fee per factory site. Three tiers:

Starter — \$99/month One camera, basic safety monitoring, up to 3 manuals in Expert-Flow.

Pro — \$199/month (*primary target*) Up to 5 cameras, full Visual-Check quality inspection, unlimited documents, multilingual output in Korean, Chinese, and English, WhatsApp and mobile alerts.

Enterprise — \$499/month Unlimited cameras, API access, on-premise deployment option, custom agent training for specific industries.

Unit economics: Infrastructure cost per factory at Pro tier: approximately \$12/month. Gross margin: ~94%. Customer acquisition through direct outreach to factory associations, government smart manufacturing programs, and system integrators.

Why SaaS works here: Factories don't buy software once — they subscribe because the AI gets more useful over time as it accumulates their specific data. An Expert-Flow agent trained on 3

years of a factory's own maintenance logs and SOPs becomes irreplaceable. That's the retention mechanism.

SLIDE 14 • ROADMAP

Phase 1 — Now to Q3 2026: Build & Compete Deliver working MVP of all three agents. Compete in and win the Qilu competition. Use competition exposure to secure first 2 pilot factory partnerships in Yantai.

Phase 2 — Q3 to Q4 2026: Launch Public SaaS launch targeting Korea and China markets simultaneously. Onboard first 50 paying factories. Integrate with WhatsApp and KakaoTalk for alert delivery. Localize fully for Chinese manufacturing context.

Phase 3 — 2027: Scale 500 factories onboarded. Add fourth agent: automated shift handover reports. Launch API marketplace for industry-specific modules. Begin Series A fundraising based on demonstrated unit economics.

The competition is Phase 1. The demo judges will see today is already built on the real infrastructure that will power the commercial product.

SLIDE 15 • WHY SHANDONG. WHY NOW. WHY US.

Why Shandong: Shandong is China's largest manufacturing province by output. It has the highest concentration of SME manufacturers of any province in China. The local government has explicitly committed to AI-driven factory transformation under the national smart manufacturing initiative. This is the single best market in the world to launch an SME-focused factory AI platform.

Why now: The open-source AI ecosystem has only become capable enough to power this platform in the last 18 months. YOLOv8, LlamaIndex, Anomalib — these tools reached production-ready maturity in 2024 and 2025. The timing is exactly right: the technology is ready, the market policy is ready, and the factories are ready to adopt.

Why us: We are an international student team at Tongmyong University — positioned between Korea and China, fluent in both cultures and both markets. Our platform is bilingual by design. We understand the SME factory owner in Yantai and the plant manager in Busan. We are not a Western company trying to localize into Asia. We are Asia building for Asia. That is our unfair advantage.

SLIDE 16 • CLOSE

Every factory deserves AI. Not just the ones that can afford Siemens.

OMNIQ gives any factory — the 60-person metal parts shop in Yantai, the textile line in Incheon, the food packaging plant in Qingdao — the same AI operations capability that today only exists inside the world's largest manufacturers.

We are not building the next enterprise software company.

We are building the platform that brings the next industrial revolution to the factories that actually need it.

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